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To cite this article: Xiyue Wang *et al* 2024 *Phys. Scr.* **99** 116010

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Salient object detection with bayesian inference based on radar and camera fusion used in UAV obstacle avoidance

RECEIVED
4 June 2024REVISED
20 August 2024ACCEPTED FOR PUBLICATION
3 September 2024PUBLISHED
11 October 2024

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E-mail: xswang@hit.edu.cn**Keywords:** salient object detection, bayesian inference, UAV obstacle avoidance, radar camera fusion**Abstract**

Most existing salient object detection methods are sensitive to background noise and rely on prior information in UAV obstacle avoidance applications despite detection methods witnessing rapid progress. In this paper, we propose an efficient framework for salient object detection based on radar-camera fusion and iterative Bayesian optimization. A rough salient object (RSO) image is first built through radar and camera calibration. Next, the RSO image is used to calculate the edge response based on the receptive field mechanism of the primary visual cortex to construct the contour image. Finally, the above two images and the 2D Gaussian distribution are jointly integrated into an iterative Bayesian optimization scheme to get the final salient object image. Different from typical detection methods, this method suppresses background noise by filtering out irrelevant pixels using fusion information. The Bayesian inference framework's detection performance is improved by precise spatial prior, consisting of optimized contour and RSO images. Experimental results indicate that the presented algorithm performs well against state-of-the-art salient object detection methods on several reference datasets in different evaluation metrics.

1. Introduction

The salient object detection technique seeks to segment the most salient target areas in images to promote the implementation of the following visual tasks, including image captioning [1], visual tracking [2], image compression [3], and target recognition [4]. Salient object detection has two important contributions to UAV obstacle avoidance research. First, the fuzzy salient detection results can assist in accurately locating obstacles [5]. Second, the accurate salient contour detection results can be used to segment the target and the background [6].

Among the sensors used for UAV target detection, the camera and radar are widely used because of their simplicity, speed, and efficiency [7]. For example, a camera-based method [8] that matches some scale-invariant key points in consecutive frames of an image is proposed to estimate the depth of the obstacle. Scannapieco *et al* [9] studied the radar-based method of multi-target tracking strategies in UAV applications. Wessendorp *et al* [10] used radar to detect objects to achieve the UAV obstacle avoidance flight. A 3D perception radar strategy [11] was proposed to perceive small UAVs hovering in various specified air positions for formation flying. However, single-sensor detection methods based on radar or cameras have limitations. Cameras have difficulty perceiving the environment in bad weather, such as rain and fog, and radars have low perception resolution. Different sensor information can be supplemented by each other, expanding the detection dimension and improving detection accuracy. Therefore, radar and camera fusion strategies have been widely used in UAV detection tasks.

The complementary information between radar and camera can improve the robustness of the target detection in the UVA field. Bigazzi *et al* [12] proposed a real-time obstacle detection system based on radar and camera to improve the detection efficiency. Vitiello *et al* proposed a radar/vision sensor fusion method [13] to track what happens by radar measurement and fusion visual detection. Huang *et al* [14] fused the camera and radar to improve the obstacle avoidance algorithm of UAVs. A robust radar and camera fusion algorithm [15] was proposed to detect and classify targets by extracting micro-Doppler features.

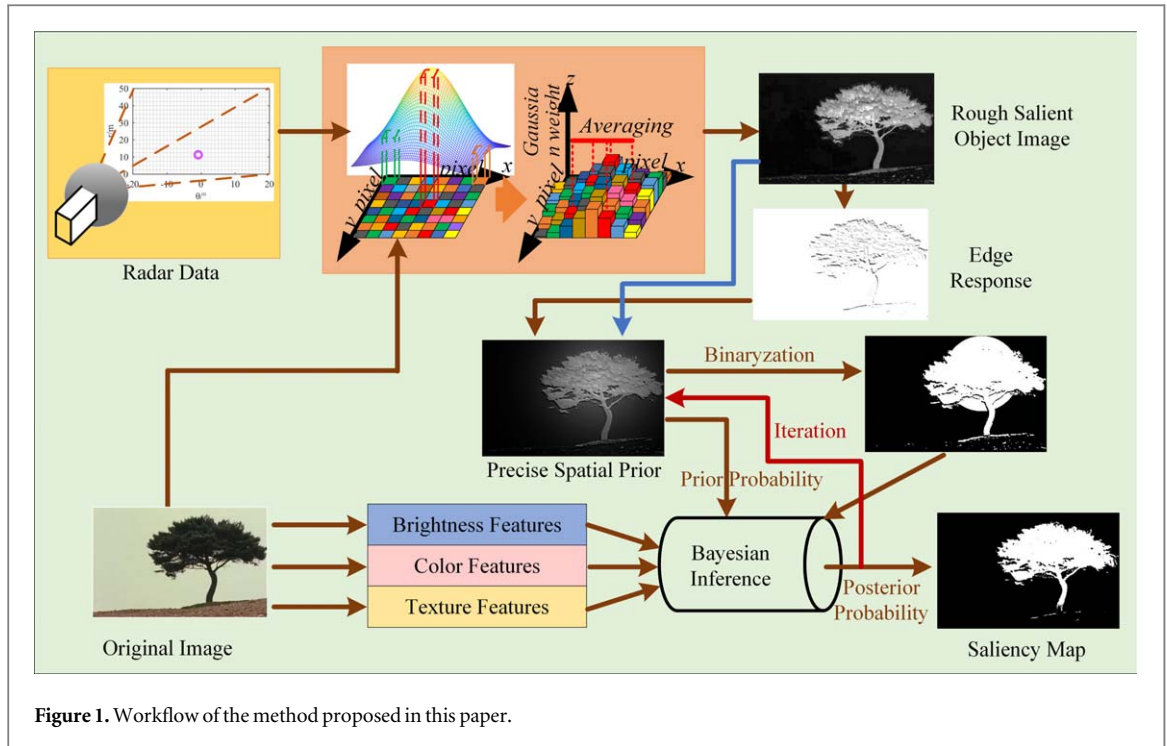
In addition, methods based on deep learning and traditional vision have been widely used at the implementation level of UAV salient target detection. The deep learning-based methods [16–18] can achieve good performance by integrating multiscale features to detect salient objects. Jian *et al* [19] designed a multiscale cascaded ResNet34 attention network. The advanced semantic feature extraction is enhanced by adding a multiscale cascading model and a channel attentiveness block. A significant improvement has been made in the accuracy and recall rate of salient object detection. Another work [20] used the ResNet101 to aggregate the features extracted by all network layers. Experimental results showed that it enhanced the aggregation of effective information. A temporal-spatial feature pyramid net [21] was proposed to extract multiscale temporal-spatial features. The temporal-spatial features from different scales were hierarchically decoded, which significantly improved the precision of saliency detection. A model [22] has been to enhance generalization performance and robustness, which incorporates image classification and saliency detection. Although the above-mentioned salient target detection methods based on deep learning have achieved good detection performance, there are still two major limitations [23]: (1) the salient objects can hardly be uniformly highlighted due to image complexity; (2) the predictions near object edges are inaccurate because of information loss caused by convolution and pooling operations.

Most of the conventional vision salient object detection algorithms [24–26] are based on local or global contrast. A salient object detection method [27] based on global contrast was proposed. Another research [28] presented a detection method based on the center-periphery Kullback-Leibler distance. Methods based on background prior information [29–31] studied salient object detection from different perspectives. Conventional methods have achieved good detection results. However, more robust methods should be developed to detect the more complex images. Bayesian inference methods are also widely used in salient object detection tasks since features such as color, space, and edges of images can be used as prior information. Bayesian inference can effectively capture complex causal relationships and dependencies in data. It can more accurately infer the saliency of objects by constructing probabilistic dependencies among factors such as color, texture, and spatial location. In addition, salient object detection usually requires consideration of multiple image features and other sensor information. Bayesian networks can easily fuse multi-source information and dynamically adjust the weight of various details according to the actual situation to improve detection accuracy and robustness. Wang *et al* [32] inferred more specific initial salient areas by a fully connected conditional random field model for improved likelihood calculation in a geodesic-weighted Bayesian framework. The method has used a salient image of existing techniques as the prior distribution. Another study on salient object detection based on Bayesian theory [33] considered color, spatial and edge information. A boundary contrast map and a contour completeness map were combined into an iterative Bayesian improved scheme to get the final salient image. A salient object detection algorithm [34] using the Bayesian edge analysis was proposed. First, the initial salient area was selected by edge analysis. The final salient image was calculated by the Bayesian theory and optimization geodesic distance. Salient object detection based on the general Bayesian framework usually has two disadvantages. First, noise in the background seriously affects detection performance. Consequently, background areas like objects are easily misdetected. Second, it relies on prior information. Thus, low-precision prior information also seriously affects the detection results.

In summary, the paper uses the traditional vision method based on the fusion of radar and camera to achieve salient object detection for UAVs. Compared with deep learning and single-sensor detection techniques based on conventional vision, the contributions of our method mainly include the following:

- (1) Enhance the robustness and accuracy of detection: Traditional vision methods are limited by light and bad weather, and their detection effect is poor in complex scenes. After fusing the radar and camera, the accurate orientation information provided by the radar can be used to help detect images, which improves the system's detection robustness and accuracy in complex environmental conditions.
- (2) Reduce noise impact: Radar can provide accurate *a priori* position information during the fusion detection. The approximate target area can be determined by combining prior information and image feature data. The influence of background noise is significantly reduced, which improves detection accuracy.
- (3) Avoiding large dataset training and low transfer rate: The fusion method proposed in the paper does not require large-scale labeled image datasets for training. In addition, the model has strong generalization ability and a high transfer rate in different data distributions since it relies on the features and information provided by physical sensors.

Specifically, we proposed a Bayesian salient object detection algorithm based on spatial prior information from the fusion of millimeter wave (MMW) radar and vision sensor. First, an RSO image is obtained through the fusion of MMW radar and vision sensor. Then, the image edge response is acquired using the primary visual cortex receptive field mechanism. Precise spatial prior information can be obtained by combining the RSO image and the edge response result. The precise spatial prior is used as the prior information of the Bayesian



framework. Then, the likelihood function is described by integrating the brightness, color, and texture features of the original image. Finally, the salient structure can be enhanced, and the background noise can be reduced by using the posterior probability of the Bayesian network to initialize the prior information for iteration. The workflow of the algorithm proposed in this paper is shown in figure 1. Our main contributions are threefold:

- (1) An RSO detection method based on radar and camera is proposed. The radar position and image pixel information are fused to filter out the background noise and choose the initial target area.
- (2) We employ the RSO image to calculate edge responses based on the primary visual cortex receptive field mechanism. The object's contour information is highlighted and refined.
- (3) The precise spatial prior consists of an RSO image, edge response contour information, and a 2D Gaussian weighted distribution. Bayesian inference is carried out by incorporating the prior information and original image features of brightness, color and texture. The salient image is iterated as new prior information to enhance the object region and reduce the background noise.

Specifically, this paper mainly includes several parts: the calculation method of the precise spatial prior was introduced in section 2. Section 3 described the structure and reasoning process of the Bayesian network. In section 4, experiments were conducted to verify the detection results on publicly available datasets and self-built millimeter wave radar and image detection datasets. Finally, some conclusions were drawn in section 5.

2. Precise spatial priors based on fused data and edge responses

2.1. Camera and radar alignment based on high-accuracy calibration method

Radar and image information are usually fused using the calibration method. The calibration can be achieved by coordinate system transformation [35] as (1).

$$\begin{aligned}
 \begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} &= \lambda \mathbf{M}_1 \mathbf{M}_2 \begin{bmatrix} X_W \\ Y_W \\ Z_W \end{bmatrix} \\
 &= \lambda \begin{bmatrix} f_x & s & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r \sin \varphi \cos \theta \\ r \cos \varphi \\ r \sin \varphi \sin \theta \\ 1 \end{bmatrix}
 \end{aligned} \tag{1}$$

Where $S(\varphi, \theta, r)$ represents the spherical coordinate value of the radar detection data. $P(x_p, y_p)$ is the pixel coordinate value of the radar detection data. M_1 is the internal parameter matrix, and M_2 is the external parameter matrix that includes a rotation matrix R and a translation matrix T . The calibration process is to find the R and the T composing the matrix M_2 .

However, the angle measurement accuracy of millimeter wave radar is generally low. The lateral offset error will be worse when the radial distance is long. It will seriously affect the calibration accuracy. Fortunately, the image contains accurate information about the object. Calibration errors can be reduced by utilizing positional information in the image. A fusion method proposed in our previous research [36] was employed, which built an artificial potential field to obtain accurate calibration points iteratively.

First, the P_t is obtained by standard calibration solution. Then, the P_r , angle measurement errors, and image are used to establish ROI. Finally, the artificial potential field is implemented inside the ROI, and iteration is repeated to complete the high-accuracy calibration.

The ROI was established based on the range of error in angle measurements of the radar. The horizontal and vertical distance offsets resulting from angle measurement errors could be calculated using (2) and (3). The Δh and Δv represent distance offsets in the horizontal and vertical directions. The $\Delta\theta$ and $\Delta\varphi$ represent the measurement errors of the azimuth and elevation.

$$\Delta h = r \sin \theta \tan \Delta\varphi \quad (2)$$

$$\Delta v = r \tan \Delta\theta \quad (3)$$

According to the camera's projection principle, the error range in angle measurement is indicated in (4) and (5). The variable f represents the camera's focal length

$$\Delta H = f \sin \theta \tan \Delta\varphi \quad (4)$$

$$\Delta V = f \tan \Delta\theta \quad (5)$$

The ROI was established as an artificial potential field by (6). Two moving points (Mp_L and Mp_R) were created at the left and right of the ROI, as shown in (7) and (8). n ($n = 1, 3, 5, 7, \dots$) is the size of the moving point. The new P was selected using the center of the line connecting the two moving points.

$$\begin{cases} x_L = x_t - \Delta H \\ x_R = x_t + \Delta H \\ y_D = y_t - \Delta V \\ y_U = y_t + \Delta V \end{cases} \quad (6)$$

$$Mp_L = \begin{bmatrix} Gray'(x_{L-1}, y_{t-n+1}) & Gray'(x_L, y_{t-n+1}) & Gray'(x_{L+1}, y_{t-n+1}) \\ \vdots & \vdots & \vdots \\ Gray'(x_{L-1}, y_t) & Gray'(x_L, y_t) & Gray'(x_{L+1}, y_t) \\ \vdots & \vdots & \vdots \\ Gray'(x_{L-1}, y_{t-n-1}) & Gray'(x_L, y_{t-n-1}) & Gray'(x_{L+1}, y_{t-n-1}) \end{bmatrix} \quad (7)$$

$$Mp_R = \begin{bmatrix} Gray'(x_{R-1}, y_{t-n+1}) & Gray'(x_R, y_{t-n+1}) & Gray'(x_{R+1}, y_{t-n+1}) \\ \vdots & \vdots & \vdots \\ Gray'(x_{R-1}, y_t) & Gray'(x_R, y_t) & Gray'(x_{R+1}, y_t) \\ \vdots & \vdots & \vdots \\ Gray'(x_{R-1}, y_{t-n-1}) & Gray'(x_R, y_{t-n-1}) & Gray'(x_{R+1}, y_{t-n-1}) \end{bmatrix} \quad (8)$$

Two moving points move opposite each other under the action of the potential field force. The attractive function of Mp_L and Mp_R is shown in (9). $Gray(Mp_L)$ and $Gray(Mp_R)$ are the grayscale pixel averages of Mp_L and Mp_R . The ε is an attractive gain coefficient.

$$U_{arr}(Mp_L) = U_{arr}(Mp_R) = \frac{1}{2} \varepsilon Gray(Mp_L) Gray(Mp_R) \quad (9)$$

The repulsive functions of Mp_L and Mp_R are mathematically given by (10) and (11). The η is a coefficient of repulsive gain.

$$U_{rep}(Mp_L) = -\frac{1}{2} \eta Gray(Mp_L)^2 \quad (10)$$

$$U_{rep}(Mp_R) = -\frac{1}{2} \eta Gray(Mp_R)^2 \quad (11)$$

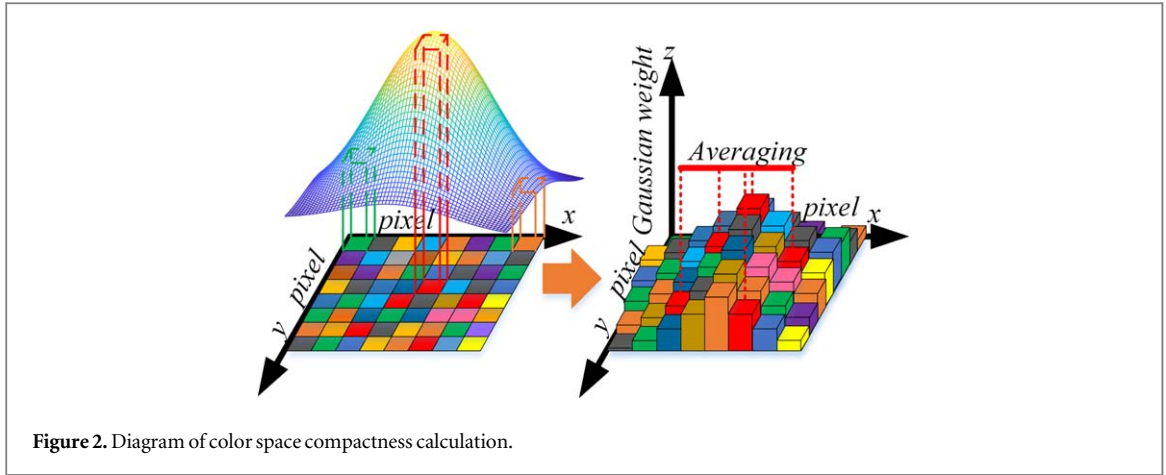


Figure 2. Diagram of color space compactness calculation.

The function that describes the state of the moving point is given by (12).

$$U(Mp_{L/R}) = U_{att}(Mp_{L/R}) + U_{rep}(Mp_{L/R}) \quad (12)$$

The evaluation condition of the motion state is given by (13).

$$\rho_{L/R} = \frac{U(Mp_{L/R})}{Gray(Mp_{L/R})} \quad (13)$$

The attractive and repulsive forces applied to the two moving points are almost equivalent when the average pixel values are similar. It indicates that both moving points are in the background or target. They will move toward each other. On the contrary, the moving point in the background will move since the attractive force is greater than the repulsive force. While the moving point in target remains stationary because the attractive force is less than the repulsive force. The process of a full motion cycle is thought of as an iteration. The iterations were repeated until the distance between the left and right moving points was less than α . Motion states are updated by (14) and (15). The symbol α represents the step size for each iteration. Finally, the average of the two moving points is the high-accuracy radar reflection point $P_r(x_r, y_r)$

$$Mp'_L(x'_L, y'_L) = \begin{cases} Mp_L(x_L + \alpha, y_L) & (\rho_L \geq 1) \\ Mp_L(x_L, y_L) & (\rho_L < 1) \end{cases} \quad (14)$$

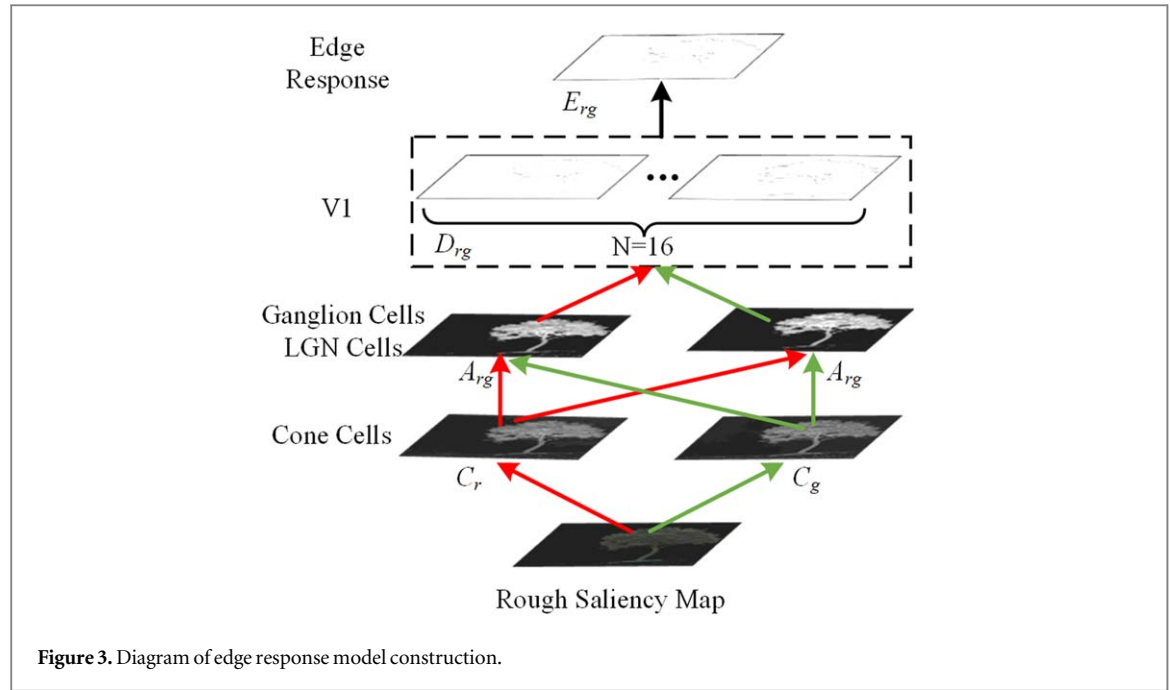
$$Mp'_R(x'_R, y'_R) = \begin{cases} Mp_R(x_R - \alpha, y_R) & (\rho_R \geq 1) \\ Mp_R(x_R, y_R) & (\rho_R < 1) \end{cases} \quad (15)$$

2.2. RSO detection based on the fusion of MMW radar and vision sensor

Radar and image information are usually fused using the calibration method. The calibration can be achieved by coordinate system transformation [35]. However, the angle measurement accuracy of millimeter wave radar is generally low. The lateral offset error will be worse when the radial distance is long. It will seriously affect the calibration accuracy. Fortunately, the image contains accurate information about the object. Calibration errors can be reduced by utilizing positional information in the image. A fusion method proposed in our previous research [36] was employed in this paper, which built an artificial potential field to obtain accurate calibration points iteratively.

The proposed RSO detection algorithm based on high-accuracy calibration points includes the following sections: First, 2D Gaussian weighting is executed with the high-accuracy calibration point as the center. Then, the Gaussian-weighted average of each color is calculated to get the color space compactness. Finally, the RSO image can be obtained by adaptively selecting a threshold to segment the object and background in the image.

The calculation of color space compactness is shown in figure 2. First, we used the 2D Gaussian function to represent the spatial range of the object. We normalized the Gaussian function value to [0,1] in the calculation. Then, the Gaussian weight corresponding to all pixel of each color were averaged. The averaged 2D Gaussian function was the color space compactness of the image. Generally, the larger a pixel's color space compactness value, the more probable it is to be considered an object. On the contrary, it is less probable to be considered an object.



The calculation of the 2D Gaussian function centered on the calibration point is shown in (16)

$$d_{\text{gauss}}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x - x_r)^2 + (y - y_r)^2}{2\sigma^2}\right) \quad (16)$$

Where, σ is a parameter related to the image size, to reasonably express the matching between image size and Gaussian function distribution, the value of σ is shown in (17). (x_r, y_r) is the pixel coordinate of the calibration point. (x, y) represents image pixels.

$$\sigma = \min(w_i, h_i)/3 \quad (17)$$

After Gaussian weighting with the calibration point as the center, the compactness of the color space can be calculated by (18)

$$d_{\text{color}_n} = \left(\sum_{(x,y) \in \text{color}_n} d_{\text{gauss}}(x, y) \right) / N_{\text{color}_n} \quad (18)$$

Where d_{color_n} is the spatial compactness of color n , N_{color_n} is the number of pixels of color n . After calculating the spatial compactness, the object and background are segmented by adaptively selecting a threshold. Finally, the RSO image can be obtained. The optimal threshold (T_{robst}) is set by (19)

$$T_{\text{robst}} = \arg_{T_t} \max \sqrt{\sum_c (\omega_c \cdot (O_{T_t}^c - B_{T_t}^c))^2} \quad (19)$$

Where $O_{T_t}^c$ and $B_{T_t}^c$ represent the average pixel value of the segmented object point set and background point set on the three color channels of R, G and B ($c \in \{r, g, b\}$). The initial weight coefficient of the three color channels is set to 1/3, which indicates that the importance of the three color channels in the initial state is equal. T_t is the search space of the threshold. The weight coefficient of each color channel can be adjusted as shown (20) when the initial optimal threshold has been estimated. The updated weight coefficient can be substituted into (19) to estimate the optimal threshold iteratively.

$$\omega_c = \frac{1}{\sum_{c \in \{r, g, b\}} \omega_c^0} |O_{T_t}^c - B_{T_t}^c| \quad (20)$$

2.3. Edge response based on the receptive field mechanism of the primary visual cortex

We have studied the perception mechanism of color information in the primary visual cortex in the biological visual system. It is applied to construct a model of visual neurons' color antagonistic receptive field properties. Finally, the model is used for the image edge detection task. The process of model construction is shown in figure 3. It includes cones [37] that sense the color channel of the input image, ganglion cells and lateral geniculate nucleus (LGN) cells [38] that sense color and brightness characteristics by single antagonistic

neurons, and the visual cortex (V1) [39] that sense the color and brightness characteristics by double antagonistic neurons.

In the cone cells, the image is decomposed into different color channels. Let $I(x, y)$ represent the input RGB format image, then the R, G, and B channel images can be represented by $I_r(x, y)$, $I_g(x, y)$ and $I_b(x, y)$. The yellow channel $I_y(x, y)$ can be formed by superposing the red and green signals. The red and green channel images construct the R-G/G-R antagonistic signal. The blue and yellow channel images construct the B-Y/Y-B antagonistic signal. The response of the cone cells for images with different color channels is shown in (21).

$$C_c(x, y, \sigma) = I_c^{rs0}(x, y) * G(x, y, \sigma), c \in \{r, g, b, y\} \quad (21)$$

Where $I_c^{rs0}(x, y)$ represents the RSO image, $G(x, y, \sigma)$ is a 2D Gaussian function representing a certain spatial range of cone cells collecting light signals, as shown in (22).

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (22)$$

The light signals processed by cones are transmitted to ganglion cells and LGN cells. Although LGN cells have larger receptive fields than ganglion cells [40], their characteristics are identical. Therefore, we combine them into one layer. The single antagonistic response model is shown in (23)

$$A_{rg}(x, y) = \mu_1 C_r(x, y, \sigma) + \mu_2 C_g(x, y, \sigma) \quad (23)$$

Where μ_1 and μ_2 represent the connection strength between cone cells and ganglion cells, which are satisfied the relationship of (24).

$$\mu_1 \cdot \mu_2 < 0, \text{ and } |\mu_1|, |\mu_2| \in [0, 1] \quad (24)$$

In V1, neurons have double antagonistic receptive fields for color and space. As shown in (25), the spatial structure of the double antagonistic receptive field was constructed using the first derivative of the 2D Gaussian function.

$$S(x, y, \sigma, \theta) = \frac{\partial g(x, y, \sigma, \theta)}{\partial x'} \quad (25)$$

Where $g(x, y, \sigma, \theta)$ represents a 2D Gaussian function considering different spatial orientations, as shown in (26) and (27).

$$g(x, y, \sigma, \theta) = \frac{1}{2\pi(k\sigma)^2} \exp\left(-\frac{x'^2 + (\gamma y')^2}{2(k\sigma)^2}\right) \quad (26)$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \cos \theta + y \sin \theta \\ -x \sin \theta + y \cos \theta \end{bmatrix} \quad (27)$$

Where, γ represents the width-length ratio of the receptive field space. θ is the spatial orientation of the receptive field. $k\sigma$ represents the spatial range of the V1 receptive field. This paper sets $k = 2$, which indicates that the receptive field of V1 is approximately twice that of ganglion and LGN cells [41].

In summary, the double antagonistic response model of the V1 can be obtained by convolving the 2D Gaussian function with different spatial orientations and the single antagonistic response model, as shown in (28) to (31).

$$D_{rg}(x, y, \sigma, \theta_i) = A_{rg}(x, y) * R_1 - A_{gr}(x, y) * R_2 \quad (28)$$

$$R_1 = F(S(x, y, \sigma, \theta_i)) \quad (29)$$

$$R_2 = F(-S(x, y, \sigma, \theta_i)) \quad (30)$$

$$F(t) = \begin{cases} t, & t > 0 \\ 0, & t < 0 \end{cases} \quad (31)$$

Where, θ_i represents N different receptive field orientations, as shown in (32).

$$\theta_i = \frac{(i-1)}{N} 2\pi, \theta_i \in [0, 2\pi) \quad (32)$$

We set $N = 16$ and took the maximum value of the double antagonistic model response in different spatial orientations as the final edge response, as shown in (33). The edge responses of other channels can be obtained through similar calculation steps.

$$E_{rg}(x, y, \sigma) = \max\{D_{rg}(x, y, \sigma, \theta_i) | i = 1, 2, \dots, N\} \quad (33)$$

2.4. Precise spatial prior

We estimated where objects in the image were possible to appear by RSO images and edge responses. A precise spatial prior was defined to represent the possibility that salient objects appear in the image at different locations, as shown in (34).

$$S_p(x, y, \sigma) = (d_T(x, y) + E_T(x, y, \sigma) + d_g(x, y))/3 \quad (34)$$

Where $d_T(x, y)$ is the RSO filtered by the optimal threshold. $d_g(x, y)$ is a 2D Gaussian function centered on the calibration point. $E_T(x, y, \sigma)$ is selected as the maximum edge response value of different channels, as shown in (35).

$$E_T(x, y, \sigma) = \max\{E_c(x, y, \sigma) | c \in \{rg, gr, by, yb\}\} \quad (35)$$

3. Salient object detection algorithm based on bayesian inference

As shown in figure 1, the presented Bayesian salient object detection system consists of prior and original image feature information. The above section has constructed the precise spatial prior information. The original image feature information mainly includes three parts. The first is the brightness feature, as shown in (36).

$$f_l(x, y) = (I_r(x, y) + I_g(x, y) + I_b(x, y))/3 \quad (36)$$

The second are color features. It contains two color channels: red-green and blue-yellow, as shown in (37) and (38).

$$f_{rg}(x, y) = I_r(x, y) - I_g(x, y) \quad (37)$$

$$f_{by}(x, y) = I_b(x, y) - (I_r(x, y) + I_g(x, y))/2 \quad (38)$$

The last is texture features. We used the original image's edge response to describe texture features, as shown in (39).

$$f_t(x, y) = E_T(x, y, \sigma) | I(x, y) = I_{original}(x, y) \quad (39)$$

We used a Bayesian inference model to combine prior information and features of the four channels. Based on Bayesian theory, the probability $p(o | (x, y))$ of the pixel (x, y) being part of the salient object is described in (40)

$$p(o | (x, y)) = \frac{p(o)p((x, y)|o)}{p(o)p((x, y)|o) + p(b)p((x, y)|b)} \quad (40)$$

Where $p(o)$ and $p(b) = 1 - p(o)$ represent the prior probability that the pixel belongs to the object and background. $p((x, y)|o)$ and $p((x, y)|b)$ are the likelihood functions of the observed salient objects sample set and the background sample set. We used the precise space prior as the original prior probability of salient objects, $p(o) = S_p(x, y, \sigma)$. The likelihood functions can be obtained from the image feature channels.

In order to calculate the likelihood function of the observed salient object sample set and background sample set, the possible object is estimated from the input image. Similarly, we binarize the spatial prior by adaptively selecting a threshold to get the estimated salient object point set and background point set. The optimal threshold is illustrated by (41).

$$T_{obbst} = \arg_{T_k} \max \sqrt{\sum_f (\omega_f^0 \cdot (O_{T_k}^f - B_{T_k}^f))^2} \quad (41)$$

Where $O_{T_k}^f$ and $B_{T_k}^f$ represent the estimated pixel average values of the object point set and the background point set on the four feature channels $f, f \in \{f_b, f_{rg}, f_{by}, f_t\}$. ω_f^0 represents the weight coefficient of the four channels. Let the initial weight coefficient of the four feature channels be 1/4, which indicates that the importance of the four feature channels is identical in the initial state. T_k is the search space of the threshold. After getting the initial salient object pixel set and background pixel set based on the optimal threshold, the weight coefficient of each feature channel is adjusted, as shown in (42).

$$\omega_f = \frac{1}{\sum_{f \in \{f_b, f_{rg}, f_{by}, f_t\}} \omega_f} |O_{T_{obbst}}^f - B_{T_{obbst}}^f| \quad (42)$$

The likelihood function of the salient object and background can be calculated based on the known set of salient object pixels and background pixels and the weight coefficient of each feature channel. To facilitate the calculation of each feature's probability distribution, we assume that each feature is independent [42]. Based on this assumption, the likelihood functions of the salient object pixel set and background pixel set are shown in (43) and (44), respectively

Algorithm 1. Implementation of Bayesian inference.

Input: $S_p(x, y, \sigma), f_l, f_{rg}, f_{by}, f_t$
Output: $p(o|(x, y))$

- 1: // Define Initial Value
- 2: Set $\omega_{f_l}^0 = \omega_{f_{rg}}^0 = \omega_{f_{by}}^0 = \omega_{f_t}^0 = 0.25$
- 3: Set $p(o)^0 = S_p(x, y, \sigma)$
- 4: Set $p(b)^0 = 1 - p(o)^0$
- 5: // Define the Number of Iterations
- 6: Set $I = I_0$
- 7: **for** $i < I$ **do**
- 8: // Initialization Parameter
- 9: Set $T = T_0$
- 10: Set $p(o)^i = p(o)^{i-1}$
- 11: Set $p(b)^i = 1 - p(o)^i$
- 12: Set $T_{obbst}^i = 0$
- 13: // Find the Optimal Threshold
- 14: **for** $t < T$ **do**
- 15: $T_{obbst}^t = \max(T_k^t)$
- 16: **if** $T_{obbst}^i < T_{obbst}^t$ **then**
- 17: $T_{obbst}^i = T_{obbst}^t$;
- 18: **else**
- 19: $T_{obbst}^i = T_{obbst}^i$;
- 20: **end if**
- 21: **end for**
- 22: // Update the Weight Coefficient
- 23: $\omega_f^i = (\omega_f^{i-1}, T_{obbst}^i)$
- 24: // Solve the Likelihood Function
- 25: $p((x, y)|o)^i = (\omega_f^i, f_l, f_{rg}, f_{by}, f_t, T_{obbst}^i)$
- 26: $p((x, y)|b)^i = (\omega_f^i, f_l, f_{rg}, f_{by}, f_t, T_{obbst}^i)$
- 27: // Calculate the Posterior Probability
- 28: $p(o|(x, y))^i = (p(o)^i, p(b)^i, p((x, y)|o)^i, p((x, y)|b)^i)$
- 29: **end for**
- 30: $p(o|(x, y)) = p(o|(x, y))^{I_0}$

$$p((x, y)|o) = \prod_{f \in \{f_l, f_{rg}, f_{by}, f_t\}} (p((x_f, y_f)|O_{T_{obbst}}))^{\omega_f} \quad (43)$$

$$p((x, y)|b) = \prod_{f \in \{f_l, f_{rg}, f_{by}, f_t\}} (p((x_f, y_f)|B_{T_{obbst}}))^{\omega_f} \quad (44)$$

Where $p((x_f, y_f)|O_{T_{obbst}})$ and $p((x_f, y_f)|B_{T_{obbst}})$ represent the distribution functions of the salient object point set under the four feature channels. The distribution function of each feature can be described by normalizing the histograms of shown in (45) and (46)

$$p((x_f, y_f)|O_{T_{obbst}}) = N(x_f^o, y_f^o) / N(O_{T_{obbst}}) \quad (45)$$

$$p((x_f, y_f)|B_{T_{obbst}}) = N(x_f^b, y_f^b) / N(B_{T_{obbst}}) \quad (46)$$

Where $N(x_f^o, y_f^o)$ is the number of pixels in the salient object point set that contain feature (x_f, y_f) . $N(O_{T_{obbst}})$ is the number of pixels in the salient object point set. Similarly, $N(x_f^b, y_f^b)$ is the number of pixels in the background point set that contain feature (x_f, y_f) . $N(B_{T_{obbst}})$ is the number of pixels in the background point set. The weight coefficient ω_f reflects the contribution of different features to the probability distribution function.

After obtaining the likelihood function and prior information, the final posterior probability can be calculated by (40). It described the salient object probability of each pixel. The prior information is initialized through the posterior object point set and the background point set, as probability, and the updated weight coefficient is the salient initialized through the initial weight coefficient to find the optimal threshold. The posterior probabilities can be estimated iteratively using Bayesian inference. The iterative results can enhance the salient structure and suppress background interference. The entire algorithm flow is shown in Algorithm 1.

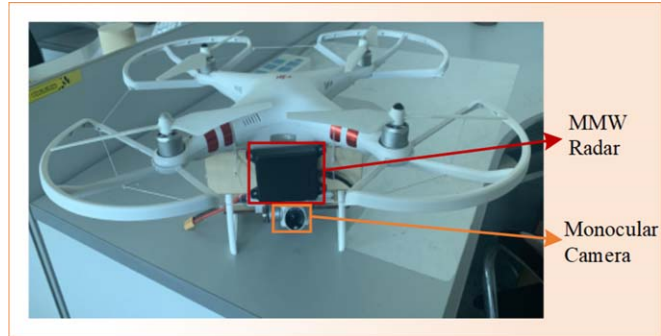


Figure 4. UAV fusion platform.

Table 1. Brand models and related parameters.

Sensor	Item	Value
Camera	Brand Model	DJI Phantom 3 Camera
	Image Sensor	1/2.3 inch CMOS
	Lens	FOV 94° 20 mm (35 mm)
	ISO Range	Video: 100–3200, Photo: 100–1600
	Shutter Speed	8 - 1/8000 s
	Resolution	4000 × 3000
	Video Stream	40 Mbps
Radar	Brand Model	NANORADAR MR
	Modulation mode	FMCW
	Ranging range	medium range mode: 0.2–80 m short range mode: 0.2–40 m
	Range resolution	medium range mode: 0.31 m short range mode: 0.17 m
	Range accuracy	medium range mode: 0.8 m short range mode: 0.10 m
	Angle range	medium range mode: $\pm 15^\circ$ short range mode: $\pm 56^\circ$
	Angle accuracy	$\pm 2^\circ$
	Speed range	-100 km h^{-1} to $+100 \text{ km h}^{-1}$
	Speed resolution	medium range mode: 1.73 km/h short range mode: 2.06 km h ⁻¹
	Speed accuracy	medium range mode: 0.87 km/h short range mode: 1.03 km h ⁻¹
	Pitch beam	medium range mode: 15.6° short range mode: 14.0°
	Azimuth beam	medium range mode: 30° short range mode: 112°

4. Simulations and experiments

In this section, we evaluated the performance of the proposed algorithm. First, the performance of the edge detection algorithm based on RSO was tested on the BSDS500 dataset [43] and a self-built dataset. Since we focused on the detection results of salient objects, we executed two operations on the BSDS500 dataset. One was to suppress the background edge information in the ground truth to highlight the edge information of the salient object. The other was to simulate the radar calibration point information of objects on the BSDS500 dataset based on the distribution of radar measurement errors on the self-built dataset. Second, the performance of the algorithm proposed and other salient object detection algorithms were compared and analyzed on DUTS [44], HKU-IS [45], SOD [46], ECSSD [47], and self-built datasets. The self-built dataset was built on a UAV platform based on the fusion of MMW radar and vision sensor. The UAV experimental platform is shown in figure 4. The brand models and related parameters of radar and camera used are shown in table 1. The edges of the objects in the self-built dataset were artificially drawn, and the contours of the salient objects were marked.

4.1. Analysis on the salient object edge detection

First, we tested several main parameters of the proposed algorithm, including the scale parameter σ representing the spatial range of the receptive field in (22) and (26), and the connection strength coefficients (μ_1, μ_2) in (23). According to the value relationship between μ_1 and μ_2 in (24), we set $\mu_1 = 1, \mu = \mu_2/\mu_1 = \mu_2$. In the experiment, different values of μ and σ were used to verify the effect of the connection strength coefficients and the receptive

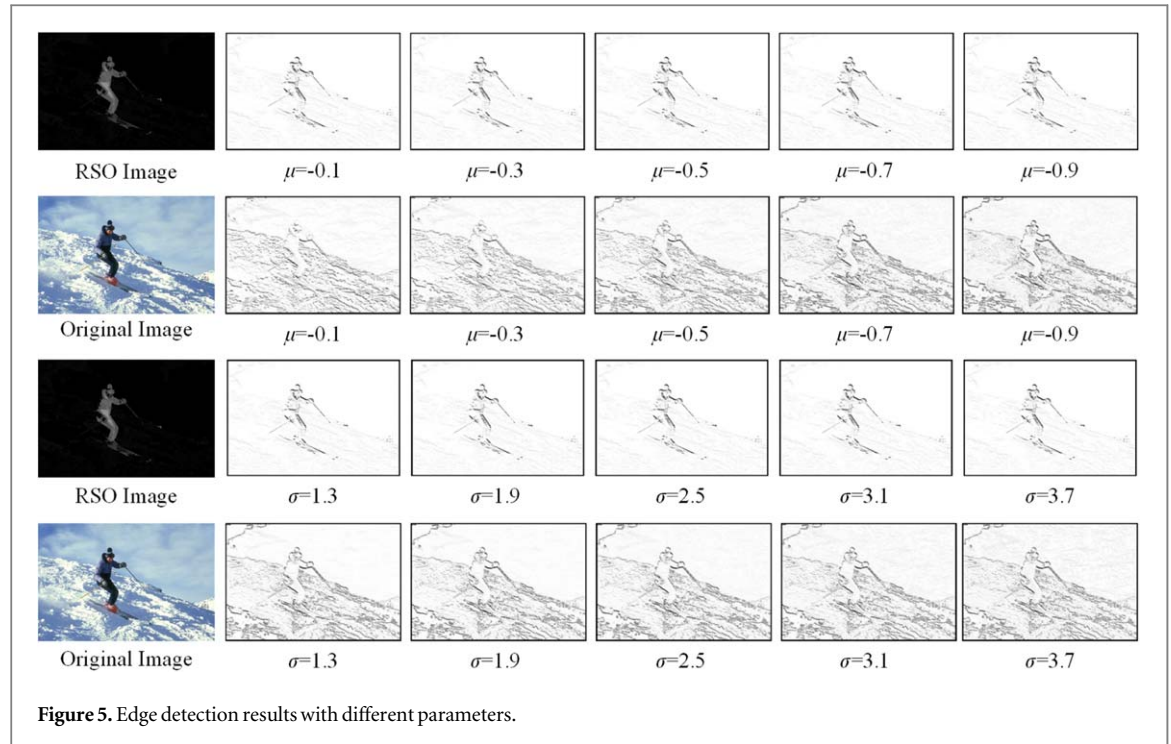


Figure 5. Edge detection results with different parameters.

field spatial range on the detection results. We detected the edge on the dataset for the original and the RSO images. The results are shown in figure 5.

We can qualitatively analyze and draw the following conclusions from figure 5. First, it can be seen that the algorithm proposed in this paper based on the RSO image has better edge detection results. It can significantly suppress some background noise. The contour information of the salient object is highlighted. Then, the response to edge brightness is gradually enhanced when the absolute value of the connection strength coefficient μ is increased. This situation is more obvious in the edge response calculation based on the original image. The edge detection results based on the RSO image are less affected by changing σ . The reason is that the RSO image has been filtered. The feature expression of color and brightness was weakened. Finally, the impact of the scale parameter on the detection results is not linear. The edge response can be improved by setting an appropriate value.

In order to quantitatively analyze the impact of algorithm performance and parameter settings on detection results, we introduced Precision and Recall to verify the detection results. The algorithm performance was evaluated by plotting the Precision–Recall (PR) curves. For each detection result, a binary image was generated by binarizing an edge response map with a fixed threshold from 0 to 255, and the PR curves were calculated by comparing the ground truth with the binary image.

First, the influence of different μ on the detection results was verified. We detected contour edges based on the original and RSO images on the same dataset. The detection results were refined using standard non-maximum suppression algorithms. The PR curves are shown in figures 6 and 7. Figure 6 is the contour edge detection result based on the original images. It can be observed that as the absolute value of μ increases, the detection performance also improves. But when $\mu = -1$ ($\mu_2 + \mu_1 = 0$), the detection performance will drop significantly. Figure 7 is the contour edge detection result based on RSO images. Similarly, the performance also shows a positive correlation trend with the absolute value of μ . However, the impact of μ changes on performance is much smaller. In addition, the detection performance also drops significantly when $\mu = -1$. It shows that the detection results will be vastly affected when the two connection weight coefficients cancel out. We take $\mu = -0.9$ as the default connection strength coefficients of the algorithm. The following text has no particular explanation, indicating that the contour edge detection results are based on $\mu = -0.9$. Second, we verified the impact of different σ on the detection results. The PR curves are shown in figures 8 and 9. Figure 8 is the contour edge detection result based on the original images. It can be analyzed that the detection performance is low when σ is small. When σ increases to 2.5, the detection performance reaches a higher value and decreases again. It shows that a suitable receptive field spatial range can greatly improve contour edge detection performance. If the receptive field spatial range is too large or too small, the contour edge detection ability will be weakened. Figure 9 is the contour edge detection result based on RSO images. The performance change trend is similar to figure 8. The difference is that the change of σ has a much smaller impact on the detection results based

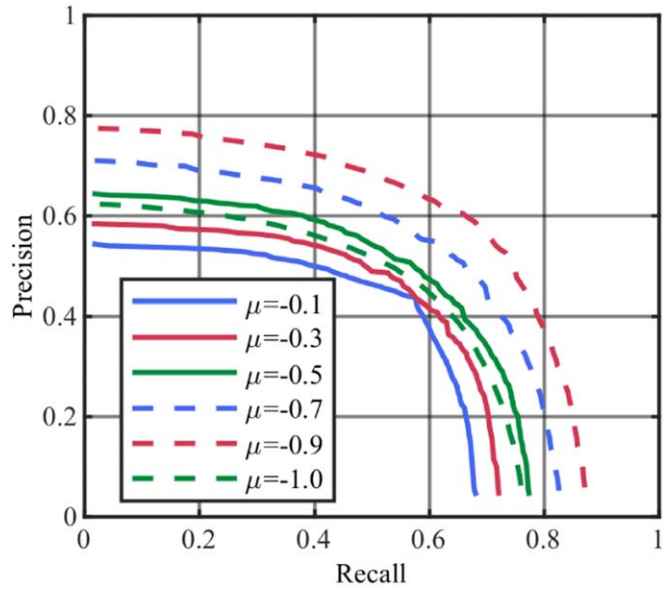


Figure 6. PR curve of edge detection results based on original images with different μ .

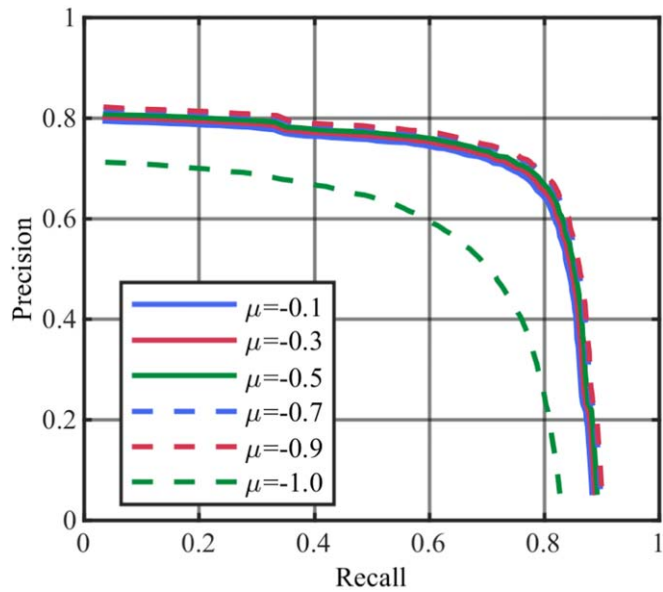


Figure 7. PR curve of edge detection results based on RSO images with different μ .

on RSO images, similar to the impact of μ on the two algorithms. We take $\sigma = 2.5$ as the default scale parameter of the algorithm. The following text has no particular explanation, indicating that the contour edge detection results are based on $\sigma = 2.5$.

Finally, we tested the contour edge detection performance based on original and RSO images on both datasets. The PR curve is shown in figure 10. It can be found from the PR curve that the presented detection algorithm performed better on both datasets. The experimental results illustrated that the RSO image filtered out the irrelevant background information of the background noise. And the contour edge lines of the salient object were enhanced. It provided a valid condition for obtaining precise prior spatial information.

4.2. Analysis on the salient object detection

We used several popular evaluation metrics to compare performance, including the PR curve, average precision (AP), F-measure, and mean absolute error (MAE). Similar to drawing the PR curve of edge detection results, a binary image was obtained by binarizing a salient object detection result with a fixed threshold from 0 to 255.

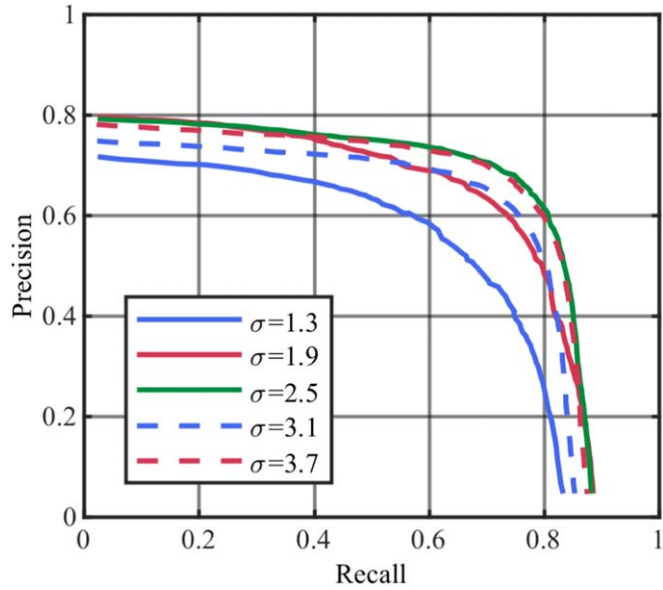


Figure 8. PR curve of edge detection results based on original images with different σ .

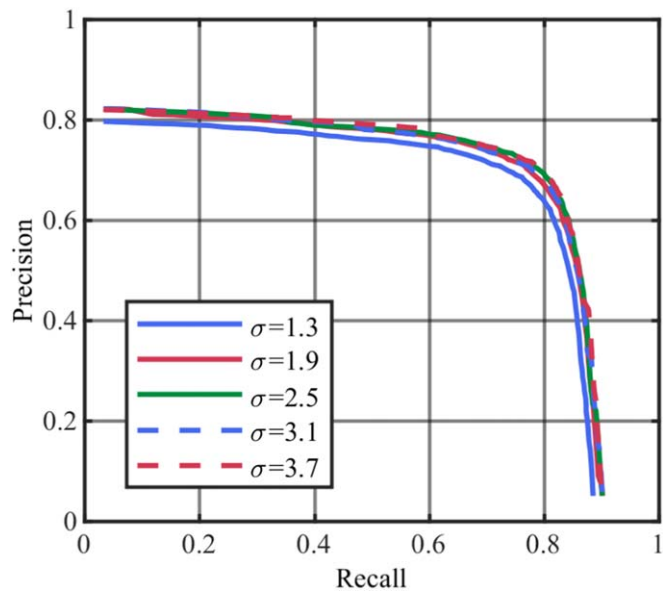


Figure 9. PR curve of edge detection results based on RSO images with different σ .

The PR curve can be generated by evaluating the binary image against the ground truth. The weighted harmonic means of precision and recall, called F-measure, was calculated to evaluate the salient object detection model comprehensively, as shown in (47).

$$F\text{-measure} = \frac{2PR}{(P + R)} \quad (47)$$

The above metrics mainly focused on the probability of correctly detecting salient pixels and ignored the impact of background pixel distribution. Therefore, MAE was used to address this issue. MAE represents the mean absolute error between the salient object detection map and ground truth.

First, we analyzed the impact of the iteration of Bayesian inference on salient object detection. The experimental results are shown in figure 11. Where, (a) to (d) represent the salient object detection maps of the first, second, third and fourth Bayesian inference. It can be concluded from (f), (g), and (h) that the suggested method converged very quickly. The PR curve has reached the optimum during the second inference. Similarly, the weight coefficients of the four feature channels also remain stable after the second inference. The AP,

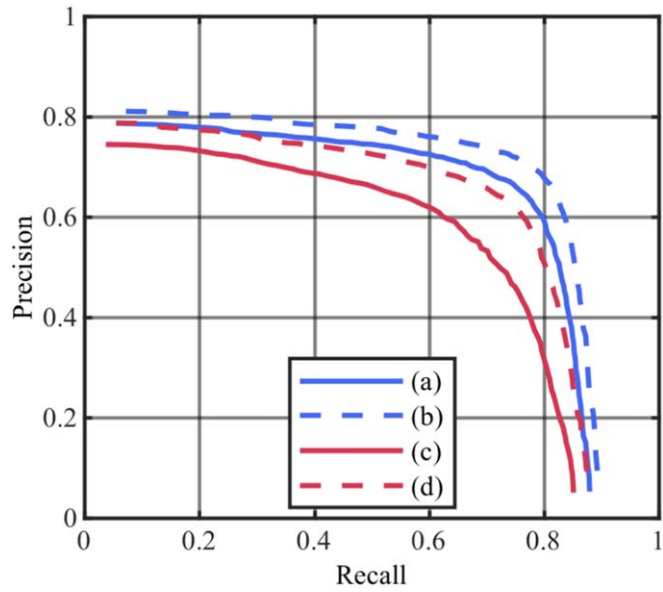


Figure 10. PR curve of edge detection results on different datasets. (a) Detection algorithm based on original images on BSDS500 dataset; (b) Detection algorithm based on RSO images on BSDS500 dataset; (c) Detection algorithm based on original images on self-built dataset; (d) Detection algorithm based on RSO images on self-built dataset.

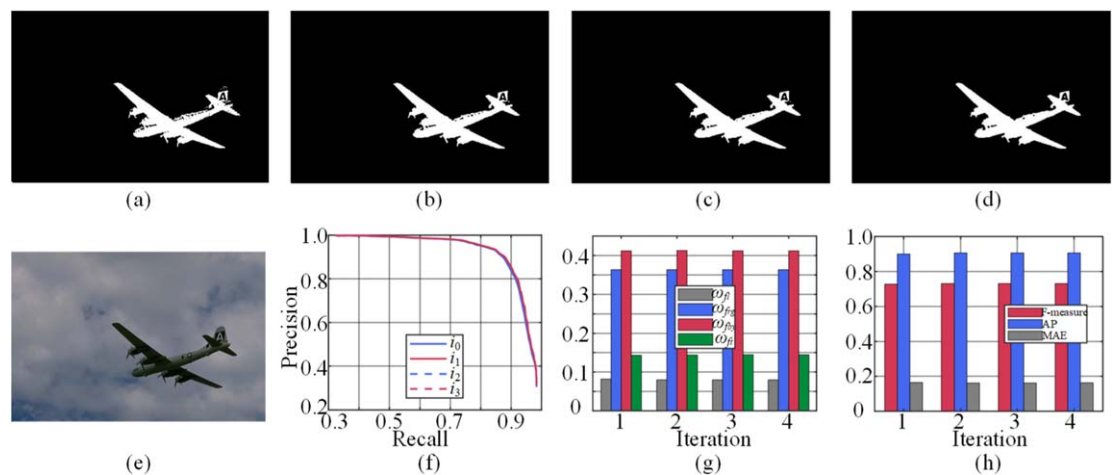


Figure 11. Detection results of different Bayesian inference times: (a) to (d) The salient object detection maps of the first, second, third and fourth Bayesian inference; (e) Input; (f) PR curves; (g) the weight coefficients of the four feature channels; (h) AP, F-measure, and MAE.

F-measure, and MAE have reached the optimum. It can be concluded that the detection performance of the presented method has reached the optimal value after two Bayesian inferences. The following text has no particular explanation, indicating that the salient object detection map results from two Bayesian inferences.

The proposed algorithm is extensively compared against 5 typical salient object detection methods, including MR [48], BFS [49], BL [50], BSCA [51] and LPS [52], on DUTS, HKU-IS, SOD, ECSSD, and self-built datasets. The experimental results are summarized in figure 12 and table 2. To verify the effectiveness of the presented algorithm, bad weather (such as low illumination, rain and fog shadows, complex backgrounds, etc) was artificially built into the self-built dataset for experiments. The results show that our algorithm ranks first among different evaluation metrics for all test datasets.

Specifically, it can be observed that our method shows the highest performance in terms of AP (0.743, 0.895, 0.864, 0.893, 0.907), F-measure (0.731, 0.881, 0.896, 0.867, 0.831) and MAE (0.135, 0.128, 0.152, 0.110, 0.092) on DUTS, HKU-IS, SOD, ECSSD and self-built datasets. Moreover, our model can reach the highest precision and recall, which means that the results generated by our model are more similar to the ground truth. It also illustrates that the developed model is more suitable for practical UAV obstacle avoidance. Compared with the experimental

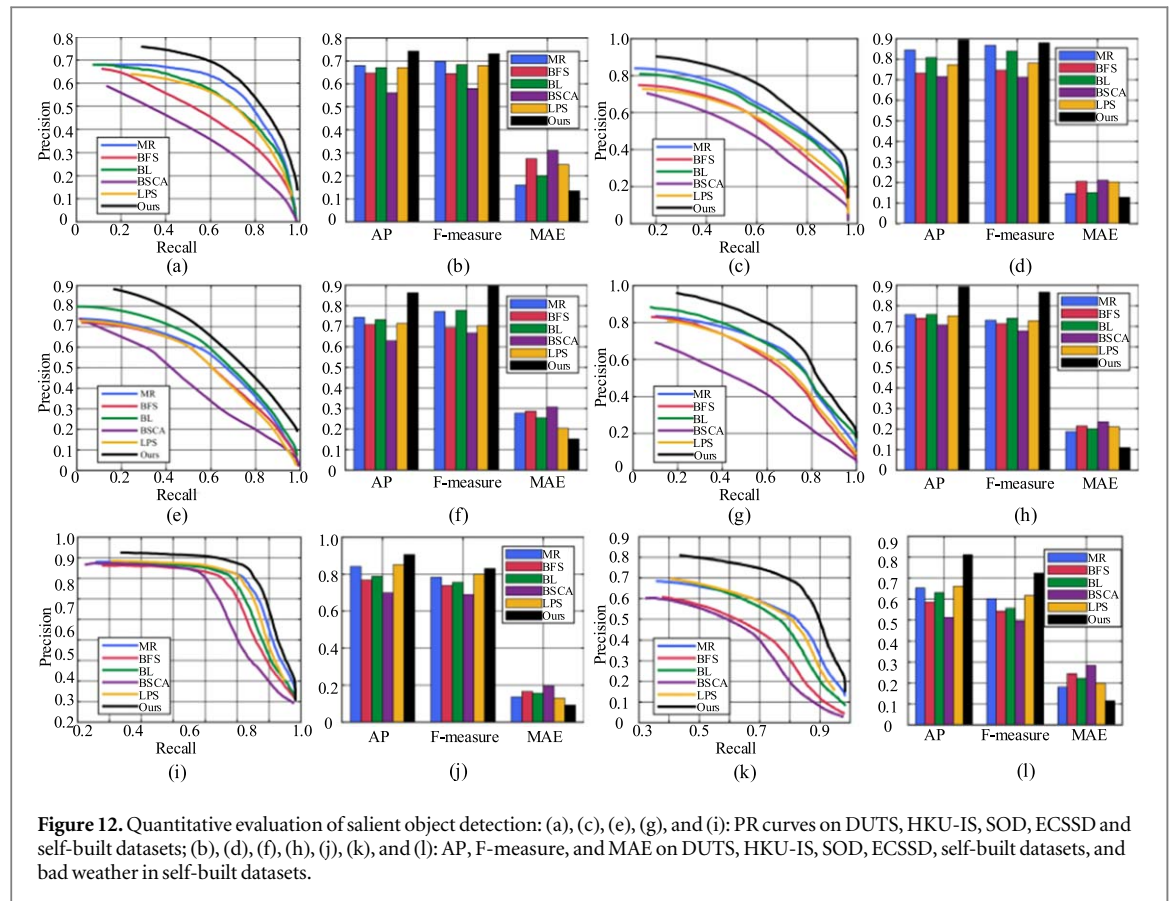


Figure 12. Quantitative evaluation of salient object detection: (a), (c), (e), (g), and (i): PR curves on DUTS, HKU-IS, SOD, ECSSD and self-built datasets; (b), (d), (f), (h), (j), (k), and (l): AP, F-measure, and MAE on DUTS, HKU-IS, SOD, ECSSD, self-built datasets, and bad weather in self-built datasets.

Table 2. Results on different datasets of AP, F-measure, and MAE.

	Method	MR	BFS	BL	BSCA	LPS	Ours
AP	DUTS	0.680	0.647	0.671	0.561	0.671	0.743
	HKU-IS	0.845	0.733	0.809	0.715	0.773	0.895
	SOD	0.744	0.709	0.733	0.629	0.714	0.864
	ECSSD	0.758	0.739	0.758	0.707	0.751	0.893
	Self-Built	0.842	0.769	0.789	0.700	0.852	0.907
	Bad Weather	0.653	0.586	0.632	0.513	0.661	0.811
F-measure	DUTS	0.697	0.644	0.683	0.580	0.680	0.731
	HKU-IS	0.868	0.747	0.840	0.712	0.782	0.881
	SOD	0.772	0.692	0.778	0.667	0.703	0.896
	ECSSD	0.730	0.713	0.739	0.677	0.727	0.867
	Self-Built	0.784	0.738	0.756	0.690	0.800	0.831
	Bad Weather	0.602	0.542	0.556	0.495	0.617	0.724
MAE	DUTS	0.160	0.275	0.200	0.310	0.250	0.135
	HKU-IS	0.147	0.206	0.151	0.212	0.203	0.128
	SOD	0.278	0.288	0.256	0.308	0.294	0.152
	ECSSD	0.188	0.216	0.200	0.236	0.212	0.110
	Self-Built	0.116	0.166	0.157	0.196	0.130	0.092
	Bad Weather	0.180	0.244	0.222	0.283	0.198	0.116

results on the self-built dataset, the detection performance in bad weather has declined. It shows that the salient object detection algorithm is seriously affected by bad weather. The performance of the salient object detection algorithm proposed in the paper is higher than that of other typical algorithms. The highest AP (0.811) and F-measure (0.724) and the lowest MAE (0.116) are achieved. The proposed fusion salient object detection algorithm is more robust under bad weather conditions. It is verified that the fusion-based rough detection method can suppress complex background noise. The target was highlighted, which was beneficial to detection. Furthermore, our model performs relatively more than the other approaches for PR curves on all datasets.

We provide some salient object detection maps of the proposed method and other 5 state-of-the-art algorithms in figure 13. It can be found from the results that the salient object detection maps obtained by our

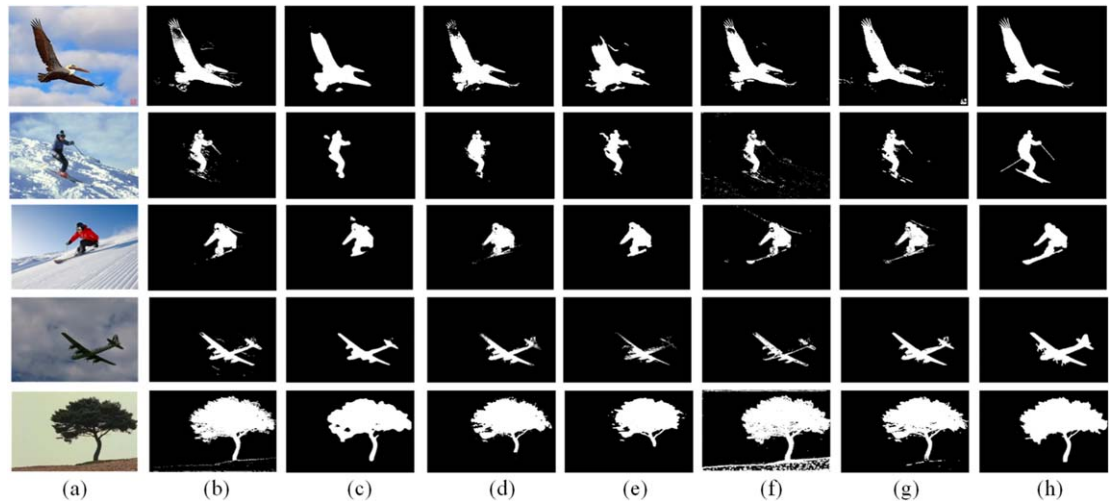


Figure 13. Comparison of our salient object detection maps with other five state-of-the-art algorithms: (a) Input; (b) MR; (c) BFS; (d) BL; (e) BSCA; (f) LPS; (g) Ours; (h) Ground-truth.

Table 3. Experimental results on FPGA.

Method	Clock	LUT (k)	BRAM
MR	100 MHz	5.3	3
BSCA	100 MHz	9.8	6
LPS	100 MHz	18.1	10
Ours	100 MHz	10.7	12

Table 4. Experimental results of running time.

Method	Processor	Dominant frequency	RAM	Times
MR	i3-2120	3.3 G	2 G	0.26 s
BSCA	i7-4790k	4.0 G	16 G	0.47 s
LPS	i7-3770	3.4 G	32 G	0.87 s
Ours	E5-2620	2.1 G	64 G	0.31 s

method have the closest similarity to the ground truth. Our method can preserve salient objects and suppress background noise well. The results indicate the effectiveness of the proposed method in highlighting salient objects and suppressing background regions.

4.3. Hardware consumption and real-time analysis

We designed different implementation experiments to verify the computation load and real-time performance of the proposed salient object detection method. First, the computation load of different algorithms is compared using the hardware resource consumption on FPGA. The FPGA used is Xilinx XC7Z020, which has 53.2k LUTs, 140 BRAMs, and other hardware resources. We selected three algorithms (MR, BSCA, and LPS) for comparison since the original articles on other algorithms did not provide quantitative indicators of computational complexity and real-time performance. The experimental results and comparisons are presented in table 3. At a clock frequency of 100 MHz, the hardware resource consumption of the proposed algorithm is 10.7k LUTs and 12 BRAMs. It indicates that the computation load of the proposed algorithm is higher than that of the MR and BSCA methods but lower than that of the LPS method.

Then, the average running time of different detection methods is shown in table 4. Experimental results show the average execution time for processing one image. It indicates that the fastest MR method only takes 0.26 s to process an image. The slowest LPS algorithm spends 0.87 s on detecting an image. The running time of our algorithm is 0.31 s. It is significantly faster than the BSCA and LPS methods and slightly lower than the MR algorithm. In summary, the proposed salient object detection algorithm has good performance in real-time, but it still needs improvement in video stream processing tasks.

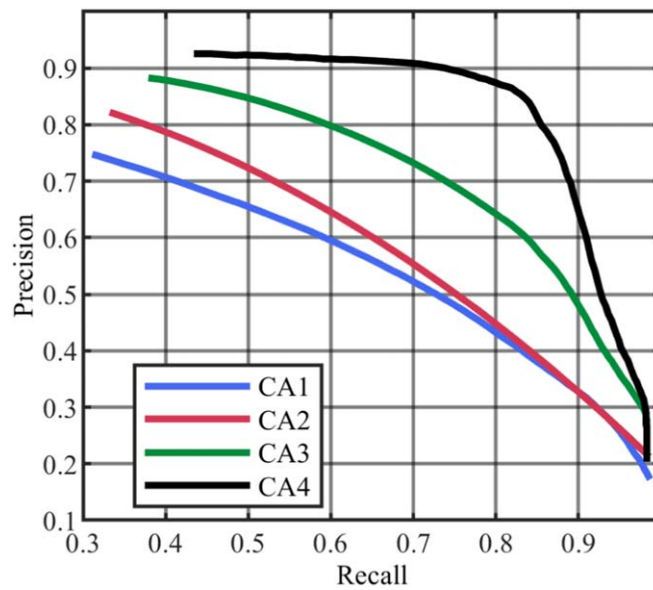


Figure 14. PR curves of different Bayesian inference algorithms.

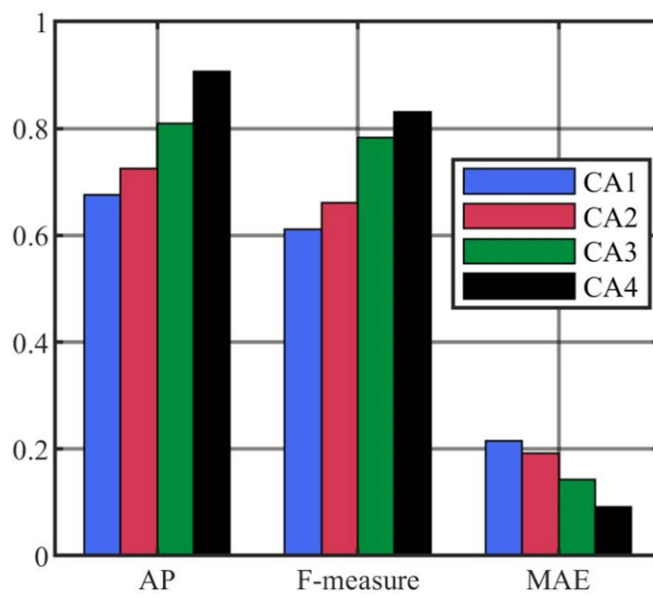


Figure 15. AP, F-measure, and MAE of different Bayesian inference algorithms.

Table 5. The prior information composition of the comparison algorithm.

Method	Prior information		
	Center-Prior	Edge response (Original image)	2D Gauss
CA1	Center-Prior	Edge response (Original image)	2D Gauss
CA2	Center-Prior	Edge response (RSO image)	2D Gauss
CA3	RSO image	Edge response (Original image)	2D Gauss
CA4	RSO image	Edge response (RSO image)	2D Gauss

4.4. Effectiveness of the fusion

To verify the effectiveness of the proposed fusion method, we designed comparative experiments for verification. The proposed Bayesian algorithm fuses the radar and camera data with prior information. The prior information consists of three parts, namely, the RSO image obtained by fusing radar and camera data, the edge response result based on the RSO image, and the 2D Gaussian distribution. We conducted comparative

Table 6. AP, F-measure, and MAE of different Bayesian inference algorithms.

Method	CA1	CA2	CA3	CA4
AP	0.676	0.726	0.810	0.907
F-measure	0.611	0.661	0.783	0.831
MAE	0.216	0.192	0.143	0.092

experiments on a self-built dataset using the classic central-prior [53] and the edge response of the original image. The different prior information for experiments is shown in table 5. CA1 uses all original image information for salient object detection. The edge response of CA2 is based on the fused RSO image. The position prior information of CA3 is based on the fused RSO image. CA4 is the fusion salient object detection method proposed in this paper. The PR curves and quantitative results are shown in figures 14, 15, and table 6.

The results show that the detection performance of the fusion-based method designed in this paper outperforms other competitive methods. The detection performance of the CA1 method using the original image information without fusion data is the worst. In addition, by comparing CA2 with CA3, it can be concluded that the detection performance based on the fusion edge response prior is better than the fusion-based spatial position prior. In summary, the comparative experiments illustrate the effectiveness of the fusion-based RSO image and edge response in suppressing background noise and highlighting salient objects.

5. Conclusion

In this work, we propose a salient object detection algorithm based on the fusion of radar and camera. Based upon the calibration of radar and images, a precise prior information acquisition mechanism, including salient position and edge, is designed. Position prior information is characterized using a rough detection algorithm based on visual features. Salient edge information can be obtained by incorporating edge responses based on the receptive field mechanism of the primary visual cortex and the position of prior information. Furthermore, we present an iterative fusion detection method based on the Bayesian inference framework. It can take advantage of precise prior information and fuse feature information into a more discriminative salient image with higher precision and recall. The proposed method achieves superior performance in terms of different evaluation metrics, compared with the state-of-the-arts on five different datasets.

However, there are still some limitations to the method. For instance, insufficient use of radar feature information affects detection accuracy. We only use the point with the highest radar reflection signal as feature information. Further research is needed on the comprehensive use of radar data (speed, angle, distance, etc). In addition, although millimeter-wave radar is not affected by light, it is difficult to image the detection object because of the limitation of detection accuracy, so the method in this paper relies heavily on image feature information. Therefore, effective detection cannot be achieved at night.

In the future, we will study effective detection methods that fuse images with comprehensive radar feature information to obtain higher salient object detection performance. We will also examine strategies to improve detection accuracy and the acquisition of object feature information in drastic conditions such as darkness (e.g., using lidar).

Acknowledgments

The authors would like to thank the Anonymous Reviewers and Editors for the profound comments. This work was partially funded by Major Scientific and Technological Innovation Project of Shandong Province of China (2021ZLGX05 and 2022ZLGX04).

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